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Insulin & Diabetes Pedigree Analysis



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INSULIN LEVELS, DIABETES PEDIGREE FUNCTION AND THEIR RELATIONSHIP WITH DIABETES OUTCOMES

Based on the Pima Indians Diabetes Dataset

ABSTRACT

This report analyzes the relationship between insulin levels, Diabetes Pedigree Function (DPF), and diabetes outcomes using the Pima Indians Diabetes Dataset. Insulin values recorded as zero were treated as missing and replaced with the median value. Exploratory data analysis was conducted to examine distributions, group differences, hereditary influences, and a composite risk score. Results indicate that diabetic individuals generally exhibit higher insulin levels, greater variability, and stronger hereditary risk. The findings demonstrate that combining physiological and genetic indicators improves diabetes risk assessment.

1. INTRODUCTION

Diabetes mellitus is one of the most common chronic diseases worldwide. Early identification of risk factors is essential for prevention and management. Two important predictors are insulin levels, which reflect metabolic function, and Diabetes Pedigree Function, which estimates hereditary susceptibility.

Objectives:

- Analyze insulin distribution after data cleaning.
- Compare insulin levels across diabetes outcomes.
- Evaluate hereditary influence using DPF.
- Develop a risk score incorporating multiple factors.
- Provide actionable healthcare insights.

2. DATASET DESCRIPTION

The Pima Indians Diabetes Dataset contains 768 observations and 9 variables. The target variable, Outcome, indicates whether a patient has diabetes.

Key Variables:

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Diabetes Pedigree Function
- Age
- Outcome

The dataset is widely used in healthcare analytics, predictive modeling, and machine learning research.

3. DATA CLEANING AND PREPARATION

A large number of insulin observations were recorded as zero. Since zero insulin values are physiologically unlikely, they were treated as missing values.

Data Preparation Steps:

- Identified missing insulin values.
- Replaced zeros with the median insulin value (125).
- Checked data consistency.
- Generated descriptive statistics.
- Prepared variables for comparative analysis.

Median imputation was preferred because insulin distribution is positively skewed and the median is less affected by outliers.

4. INSULIN DISTRIBUTION ANALYSIS

The insulin distribution is positively skewed. Most individuals have moderate insulin values, while a smaller number show extremely high measurements.

Key Findings:

- Majority of observations cluster around moderate insulin levels.
- Several extreme values indicate possible insulin resistance.
- Distribution is not normally distributed.
- Variability increases among diabetic individuals.

Interpretation:

Higher insulin variability may indicate impaired metabolic regulation and increased diabetes risk.

5. OUTCOME-BASED COMPARISON

Comparison between diabetic and non-diabetic groups reveals significant differences.

Observations:

- Diabetic patients exhibit higher average insulin levels.
- Variability is greater among diabetic patients.
- Elevated insulin may be associated with insulin resistance.
- Outcome groups show distinct metabolic profiles.

Healthcare Implication:

Patients with abnormal insulin levels should receive additional monitoring and preventive screening.

6. DIABETES PEDIGREE FUNCTION ANALYSIS

Diabetes Pedigree Function (DPF) measures hereditary diabetes risk based on family history.

Key Findings:

- Diabetic individuals have higher average DPF values.
- Stronger genetic predisposition is associated with diabetes outcomes.
- DPF complements physiological indicators such as glucose and insulin.
- Family history remains an important screening factor.

Interpretation:

Genetic predisposition significantly contributes to diabetes susceptibility and should be considered alongside clinical indicators.

7. RISK SCORE DEVELOPMENT

A composite risk score was developed:

$$\text{Risk Score} = (\text{Glucose} \times 0.3) + (\text{BMI} \times 0.1) + (\text{DPF} \times 0.1) + (\text{Age} \times 0.02)$$

Benefits:

- Combines metabolic and hereditary information.
- Simplifies risk assessment.
- Helps identify high-risk individuals.
- Supports healthcare decision-making.

Individuals with higher risk scores were more likely to belong to the diabetic category.

8. RESULTS AND DISCUSSION

The analysis demonstrates that insulin levels and hereditary factors are both important predictors of diabetes. Neither factor alone completely explains diabetes outcomes; however, together they provide a stronger assessment framework.

Major Insights:

- Insulin variability is higher among diabetic patients.
- DPF values are consistently higher for diabetic individuals.

- Risk scores successfully distinguish high-risk populations.
- Combined indicators improve predictive capability.

9. CONCLUSION AND RECOMMENDATIONS

Conclusions:

- Insulin and DPF are meaningful diabetes risk indicators.
- Hereditary and metabolic factors should be evaluated together.
- Composite risk scoring improves screening effectiveness.

Recommendations:

- Include family-history assessment during routine screening.
- Monitor abnormal insulin values closely.
- Encourage early intervention programs for high-risk groups.
- Apply machine learning techniques in future studies for improved prediction accuracy.

STATISTICAL SUMMARY TABLE

Metric	Diabetic	Non-Diabetic
Average Insulin	164.70	127.79
Average DPF	0.550	0.430

Summary of Findings:

- Diabetic individuals show substantially higher average insulin levels.
- Higher DPF values indicate stronger hereditary risk.
- Combined physiological and genetic indicators improve risk prediction.
- Risk scoring can support healthcare screening and prioritization.

References:

1. Pima Indians Diabetes Dataset.
2. Healthcare Analytics Literature.
3. Exploratory Data Analysis Methodologies.